MEDGRID SENTINEL

The generator works. The fuel doesn't come. A fuel-aware brain for hospital microgrids that turns every litre the sun saves into hours of critical care.

Team: Jackson Sixtus Abetianbe (solo entry, permitted by Handbook s8) · **Brief #13** Power for a city the grid can't protect · **Track 2** Digital Solutions & AI · **Setting B** Non-residential (healthcare) · Anneru IAS Global Energy Hackathon, 30 Oct - 1 Nov 2026

Community & target user - who, and who pays

Referral hospitals in cities whose transmission line is repeatedly cut - instantiated on **Maiduguri, Borno State** (Gombe-Damaturu-Maiduguri 330 kV line down 22 Sep - 31 Oct 2024; ~2 months in 2021). **User:** the works engineer and duty manager. **Beneficiaries:** ICU, theatre, maternity, cold-chain and emergency patients (250 beds = 91,250 bed-days/yr). **Payer:** hospital management, or an energy-service company (ESCO) repaid from diesel no longer bought.

The problem - what fails today, and why

Backup is sized in kW, but a line outage is an energy-logistics crisis: diesel need rises ~4x (3,360 L/week in a normal year, simulated) while deliveries and budgets stay flat. In Dec 2022 two Maiduguri hospitals sat dark for days with working generators and empty tanks (ICIR). Only 50% of hospitals in sub-Saharan Africa report reliable electricity (WHO et al., 2023). **Root cause:** no one decides, in advance and quantitatively, which loads the remaining litres protect.

Proposed solution - what we built and how it works

A hospital-microgrid digital twin (15-min steps, 16 tiered circuits, PV, LFP storage, generator, diesel tank and resupply chain) with a **predictive controller**: (1) a fuel-runway governor that compares diesel + battery + forecast solar with each care tier's energy need until the next expected delivery and defers the least critical loads first; (2) a clinical ride-through reserve so a failed generator never darkens the ICU; (3) efficient generator cycling. Offline web app + SMS/USSD alerts.

Proof of Concept built in the 48 hours

Replay of the real 39-day outage on NASA POWER measured weather, four controllers on identical hardware/fuel; 300-trial Monte Carlo with NREL generator-failure data; full-year economics; solar forecast scored on held-out 2024; 89 automated tests; reproducible with `npm run results`.

Brief metric (39-day outage, 6,000 L/week diesel)	Status quo (diesel)	Same PV+battery, fixed rules	MedGrid Sentinel
Critical-service hours maintained (Tier 1+2)	449 h	757 h	**937 / 937 h**
Diesel litres (litres avoided)	39,000 L	39,000 L	**36,159 L (-2,841 L)**
Life-critical Tier-1 breaks · empty-tank hours	12 · 489 h	21 · 232 h	**0 · 0 h**
P(Tier-1 powered 14 days), Monte Carlo	0%	3%	**78%**
Full measured year: diesel · CO2 · Tier-1 transfer breaks	230,844 L · 619 t · 259	-	**-91,654 L (-40%) · -246 t · 0**

Illustrative 250-bed reference hospital; simulation, not field-measured performance. Real data: NASA POWER, NREL/TP-5C00-76553, NERC Band A tariff, Oct-2026 diesel N1,950/L.

Estimated cost & who pays - can the lowest-income users afford it?

Reference design (optimizer-selected) 600 kWp PV + 600 kWh LFP + 510 kW PCS + Sentinel edge: **$704,415 (N940.4M)**. Savings N311.2M/yr -> payback **3.3 yr**, IRR 30%, 20-yr NPV N553.2M at 18%. **Who pays:** under EaaS the ESCO owns and maintains; the hospital pays N20.7M/month (80% of measured savings), zero CapEx, and keeps 20% from month one (provider IRR 24%). Energy cost per bed-day falls N8,940 -> N8,258; nothing is passed to patients. No grant assumed.

6-18-month deployment path - what gets built, by whom

M1-3 metering and circuit audit at one Maiduguri-class hospital; Sentinel in shadow mode on real data. M4-6 engineering design (IEC 62446/62619/62109 equipment, NEMSA inspection) and EaaS term sheet with a licensed installer. M7-12 install; Sentinel dispatches; SMS alerts live; measured-vs-predicted reported monthly. M13-18 sites 2-3 (Kano, Abuja).

How it scales - what changes in a second community or country

Only configuration: measured weather (same NASA POWER pipeline), load profile, tariff, feeder regime, standards. Fleet screen of 10 site-years (250- and 80-bed hospitals): payback 2.9-4.9 yr in Abuja, Kano, Maiduguri, Port Harcourt; 5.9-6.2 yr in Lagos (better grid, weaker sun) - the tool says where it pays. Fleet procurement + standard design: assumed 6% cost drop per doubling (hypothesis).

Biggest risk & how we will test it early

Modelled loads and generator condition may differ from the real hospital. Test: 3 months of shadow-mode operation on circuit-level metering before any capital; proceed only if daily-energy prediction error is within +/-15%. Secondary: theft/vandalism of panels - on-campus siting and insurance priced into the EaaS fee.

Prior-work disclosure

Nothing existed before 30 Oct 2026 12:00 PM ET. Day 1 (30 Oct) first version built with the ZCode AI coding agent; Day 2 (31 Oct) engine and product rebuilt with Claude Code (Anthropic). Open-source: Next.js, React, Recharts, Zustand, Vitest, pptxgenjs, ReportLab. Data: NASA POWER. Full statement: DISCLOSURE_STATEMENT.pdf; git history is public.

Live demo: https://jae3y.github.io/medgrid-sentinel/ · **Code & data:** github.com/Jae3y/medgrid-sentinel